

Chemistry

CYL1010



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Contents

Title	Chemistry	Number	CYL1010
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. 1 st Year	Type	Compulsory
Prerequisite	Nil		
Objectives 1. Relate the fundamental principles of chemistry with practical problems encountered for engineers and place the emphasis on problem-solving 2. Provide knowledge of engineering materials metals, composites, polymers, cement, steel, fuels, batteries and drugs. Learning Outcomes 1. Understanding the behavior of matter and materials using fundamental knowledge of their nature. 2. Understanding the basic concepts and implementation of the same in advanced and interdisciplinary areas. 3. Predict potential complications from combining various chemicals or metals in an engineering setting. Contents <i>Atomic and Molecular Structure [6 Lectures]:</i> particle in a box, angular momentum, wavefunction, multielectron atoms, homonuclear and heteronuclear diatomic molecules, non-covalent interactions <i>Solid State Chemistry [6 Lectures]:</i> Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties. <i>Surface Chemistry[3 Lectures]:</i> Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension. <i>Electrochemical Systems [5 Lectures]:</i> Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion <i>Basic Spectroscopy [7 Lectures]:</i> Selection Rules, Fundamentals FTIR, Raman, UV-visible and Fluorescence spectroscopy <i>Transition Metal Chemistry [5 Lectures]:</i> Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry <i>Organic Chemistry [8 Lectures]:</i> Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies. Textbooks 1. Silberberg, M., <i>Chemistry: The Molecular Nature of Matter and Change</i>, 6th Edition, McGraw Hill Education. 2. West, A.R., (2015), <i>Solid State Chemistry and Its Applications</i>, 2nd edition, John Wiley & Sons. 3. J. D. Lee, <i>Concise Inorganic Chemistry</i>, 5th Edition. 4. Atkins, <i>Physical Chemistry</i>, 11th Edition. Reference Books 1. McMurry, J. E. & Fay, R. C. <i>Chemistry</i>, 5th Edition, Pearson. 2. Jonathan Clayden, Nick Greeves, Stuart Warren, <i>Organic Chemistry</i>, 2nd Edition. 3. F. Albert Cotton, Geoffrey Wilkinson, <i>Basic Inorganic Chemistry</i>, Wiley Student Edition. Self Learning Material 1. https://nptel.ac.in/downloads/122101001/			

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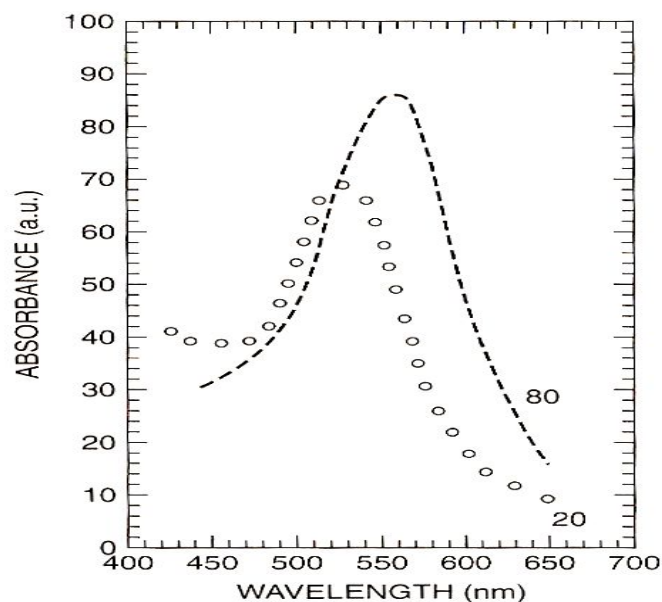
- **Surface Chemistry[3 Lectures]:** Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension.
- **Solid State Chemistry [6 Lectures]:** Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties.
- **Electrochemical Systems [5 Lectures]:** Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion

Introduction to nanomaterials based on dimensionality and properties



A piece of gold is golden in color. A colloid of golden nanoparticle is ruby red in color

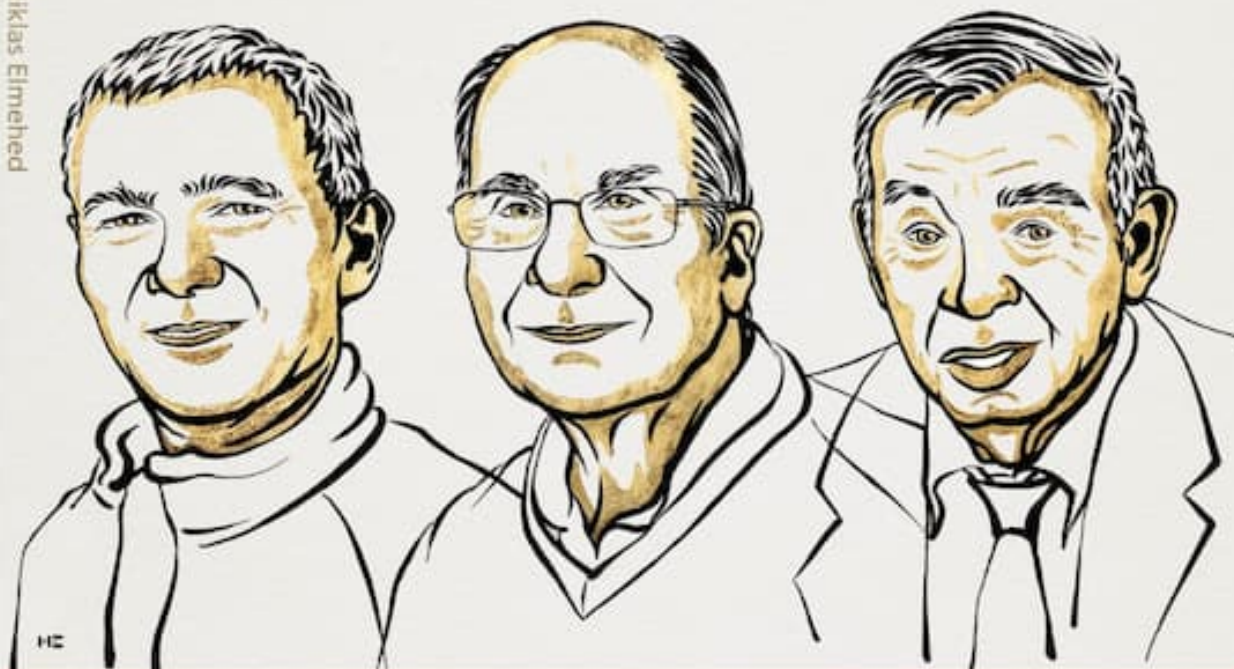
Absorption spectra of spherical Au nanoparticles



The gold we are discovering:

Material properties (such as optical Absorption, shown here) change with the size of the gold nanoparticle.

THE NOBEL PRIZE IN CHEMISTRY 2023



Moungi G.
Bawendi

Louis E.
Brus

Alexei I.
Ekimov

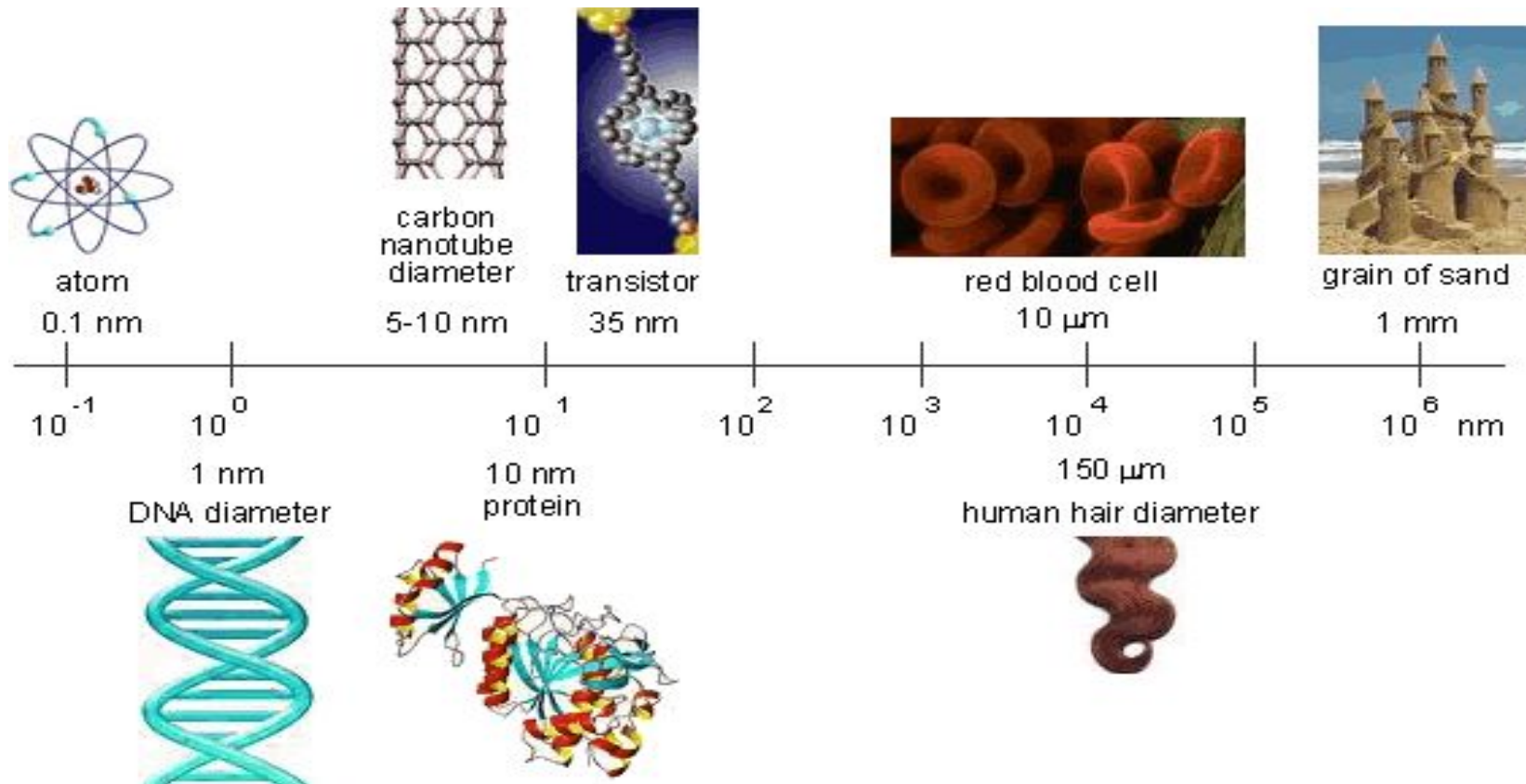
"for the discovery and synthesis of quantum dots"

THE ROYAL SWEDISH ACADEMY OF SCIENCES

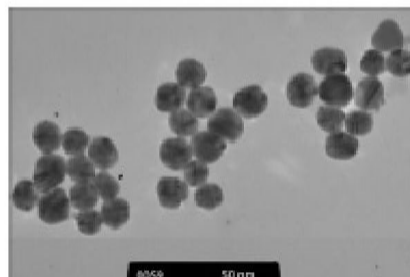
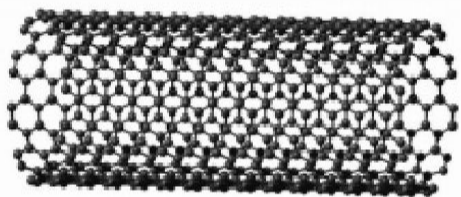
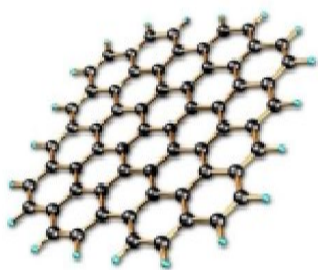
- Mounqi G. Bawendi, Louis E. Brus and Alexei I. Ekimov “for the discovery and synthesis of quantum dots”. The Nobel Prize in Chemistry 2023 rewards the discovery and development of quantum dots, nanoparticles so tiny that their size determines their properties.
- These smallest components of nanotechnology now spread their light from televisions and LED lamps, and can also guide surgeons when they remove tumour tissue, among many other things. The Nobel Prize amount for 2023 is set at Swedish kronor (SEK) 11.0 million per full Nobel Prize.
- Quantum dots now illuminate computer monitors and television screens based on QLED technology. They also add nuance to the light of some LED lamps, and biochemists and doctors use them to map biological tissue.
- Quantum dots are thus bringing the greatest benefit to humankind. Researchers believe that in the future they could contribute to flexible electronics, tiny sensors, thinner solar cells and encrypted quantum communication – so we have just started exploring the potential of these tiny particles.

Nanoscale

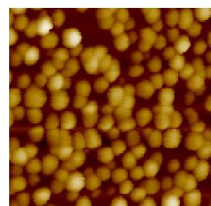
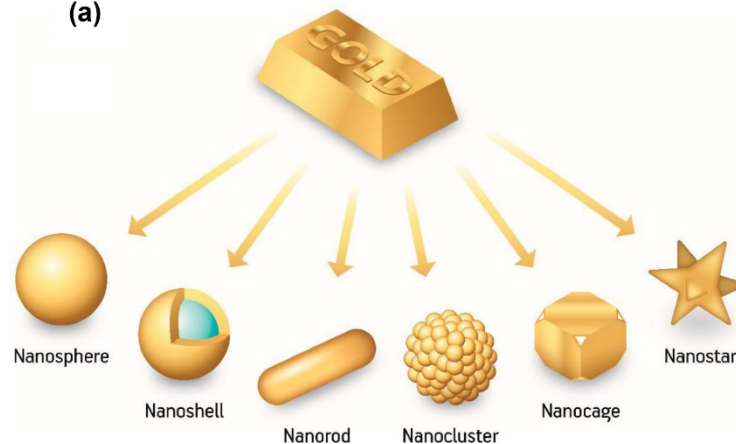
At this scale physical and chemical properties of materials differ significantly from those at a larger scale



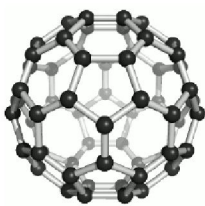
❖ Nanomaterial: refers to the matter whose length scale, in any dimension is approximately 1 to 100 nanometers



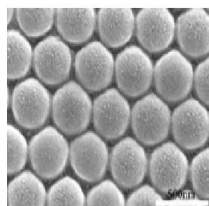
(a)



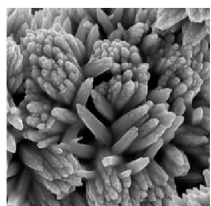
Au nanoparticle



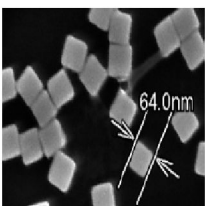
Buckminsterfullerene



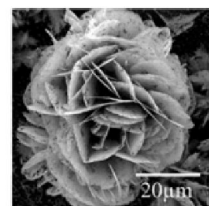
FePt nanoparticle



Titanium nanoflower



Silver nanocubes



SnO2 nanoflower



MACRO



NANO

A piece of gold is golden in color. A colloid of golden nanoparticle is ruby red in color

- The science and technology which deals with the particles in size between 1 to 100 nm is known as nano science and nano technology.

$$1 \text{ nanometer (nm)} = 10^{-9} \text{ m}$$

- **Nanoscience** is the study of phenomena and manipulation of materials at atomic, molecular and macromolecular scales, where properties differ significantly from those at a larger scale.
- **Nanotechnologies** are the design, characterisation, production and application of structures, devices and systems by controlling shape and size at nanometre scale.

Nanomaterials

“any material that has unique or novel properties, due to the nanoscale (nano meter-scale) structuring, formed by incorporation or structuring of nanoparticles”

“materials which are characterized by an ultra fine grain size ($< 100 \text{ nm}$) or by a dimensionality limited to 100 nm ”

What happens on reducing the dimension of materials in nanoscale from the bulk scale.

- High Surface area to volume ratio
- Quantum confinement (size dependent optoelectrical properties)
- Wave particle duality. The electrons exhibit wave behavior

Advantages of nano-sized Regime

- *Large surface area/volume ratio exhibited by nanomaterials □ very high surface reactivity with the surrounding surface, Eg. catalysis or sensor applications*
- *Ability of varying their fundamental properties (e.g., magnetization, optical properties (color), melting point, hardness, etc.), relative to bulk materials without a change in chemical composition.*

Surface Area : Volume Ratio

- This gives an approximate surface area to volume ratio of $>10^7:1$ which is significantly larger than a macro sized particle.
- As the surface area to volume ratio increases so does the percentage of atoms at the surface and surface forces become more dominant.
- Generally accepted material properties are derived from the bulk, where the percentage of atoms at the surface is miniscule(extremely small). These properties change at the nanoscale.

Surface Area: Volume Ratio

Nanoparticles exhibit unique properties due to their high surface area to volume ratio.

A spherical particle has a diameter (D) of 100nm.

Calculate the volume (V) and surface area (SA)

$$V = \frac{4}{3} \pi r^3 = \frac{\pi D^3}{6}$$

$$V = \frac{\pi(100 \times 10^{-9})^3}{6}$$

$$V = 5.24 \times 10^{-22} \text{m}^3$$

$$SA = 4\pi r^2 = \pi D^2$$

$$SA = \pi(100 \times 10^{-9})^2$$

$$SA = 3.141 \times 10^{-14} \text{m}^2$$

Surface Area: Volume Ratio

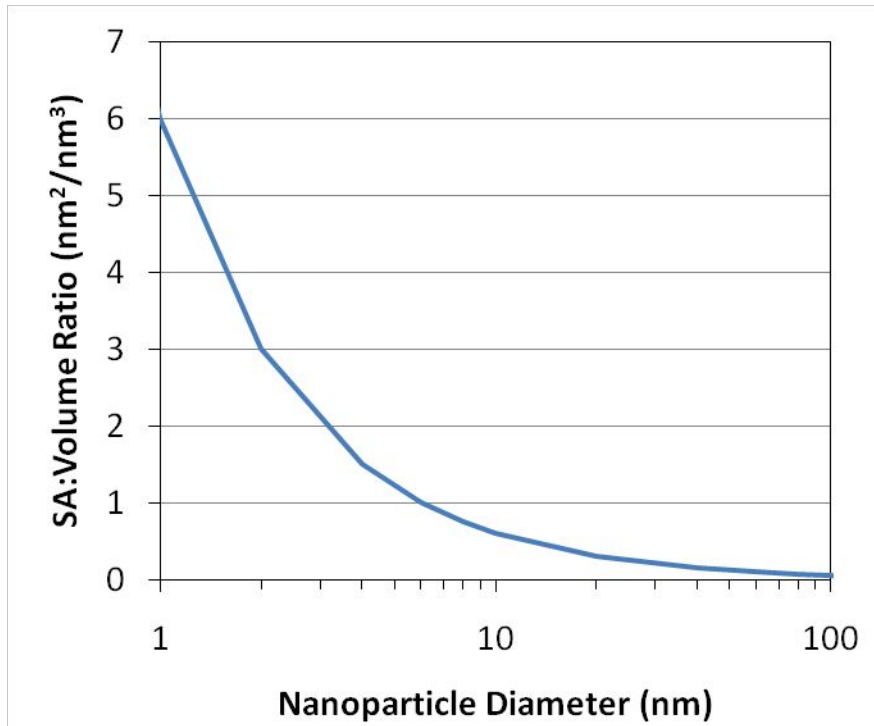
Some example calculations for volume and surface area of nanoparticles.
These calculations use nm as unit of length.

Nanoparticle Diameter (nm)	Nanoparticle Diameter (um)	Volume (nm ³)	Surface Area (nm ²)	SA:Vol Ratio (nm ² /nm ³)
1	0.001	0.524	3.14	6
10	0.01	524	314	0.6
100	0.1	523598	31416	0.06
1000	1	5.24E+08	3.14E+06	0.006
10000	10	5.24E+11	3.14E+08	0.0006
100000	100	5.24E+14	3.14E+10	0.00006
1000000	1000	5.24E+17	3.14E+12	0.000006

Why is Large Surface Area Important?

The large surface area to volume ratio of nanoparticles opens many possibilities for creating new materials and facilitating chemical processes. In conventional materials, most of the atoms are not at a surface; they form the bulk of the material. In nanomaterials, this bulk does not exist. Indeed, nanotechnology is often concerned with single layers of atoms on surfaces. Materials with this property are unique. For example, they can serve as very potent catalysts or be applied in thin films to serve as thermal barriers or to improve wear resistance of materials.

Surface Area: Volume Ratio



In this graph:

SA = nm²

Vol = nm³

SA:Vol Ratio = nm²/nm³

The ratio increases dramatically when the nanoparticle diameter drops below about 100 nm

What happens on reducing the dimension of materials in nanoscale from the bulk scale.

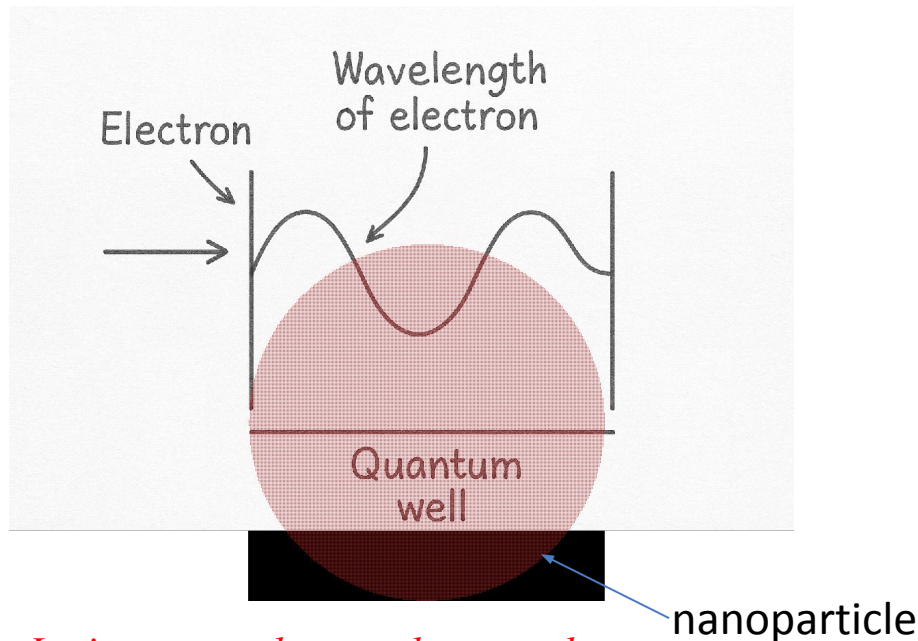
- High Surface area to volume ratio
- **Quantum confinement (size dependent optoelectrical properties)**
- **Wave particle duality. The electrons exhibit wave behavior**

To be discussed now

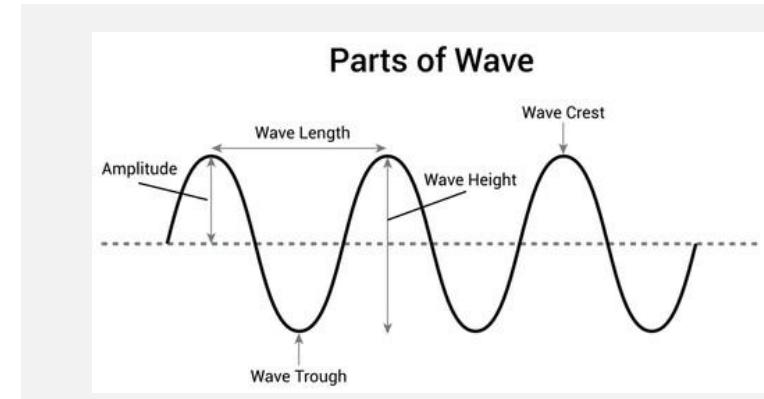


Discussion on quantum dots and quantum confinement

When a particle is small enough, less than its electron's [de Broglie wavelength](#) (around a few nanometers), the [particle acts as a potential well](#) that restricts the electron's movement, leading to the [quantum confinement effect](#). This [confinement changes the electron's energy into discrete levels rather than a continuous band](#), [increasing the band gap and affecting the material's electronic and optical properties, such as absorption and emission](#)



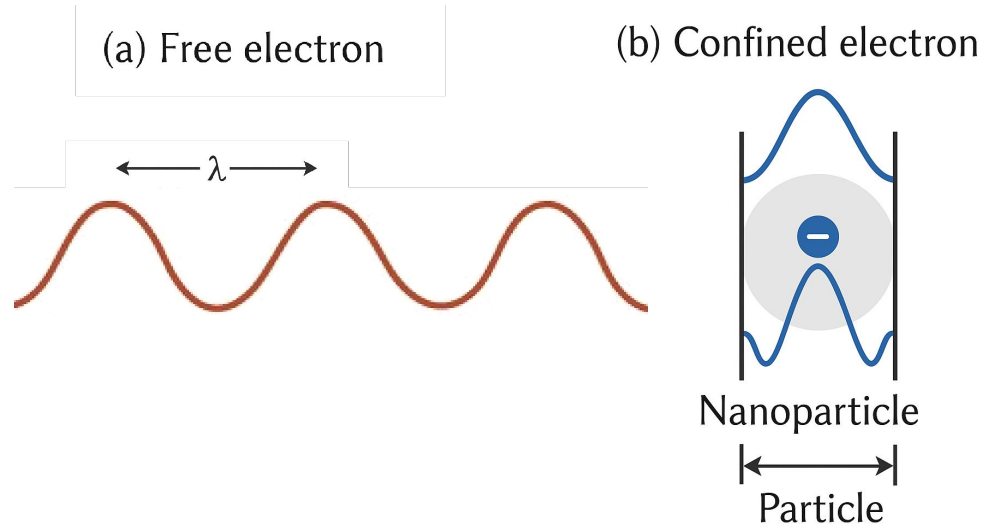
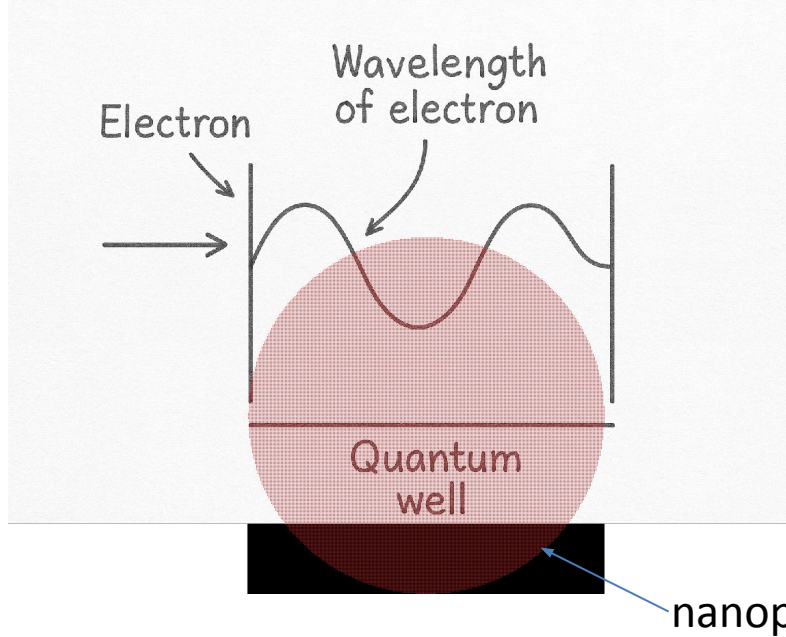
Let's assume that you have made a nanoparticle which has size (r) comparable or less than the wavelength of the electron.



An electron is not just a particle; it also has a **de Broglie wavelength**

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

where h is Planck's constant, m is electron mass, and E is its kinetic energy.



Particle size comparable to or smaller than electron wavelength

- Imagine an electron moving freely in a material (like a metal or semiconductor).
- If the particle (say a nanoparticle or quantum dot) has a size **larger than the wavelength**, the electron can fit inside with many oscillations of its wavefunction → it behaves almost like a free electron.
- But if the particle size becomes **comparable to or smaller than the wavelength**, the electron's wave cannot oscillate freely — it gets **confined**. *Because the box is smaller than the electron's natural "wavy" size (its de Broglie wavelength), the electron cannot spread out beyond the box.*

Restriction of Movement: *When the size of a particle becomes comparable to or smaller than the electron's wavelength, the particle acts as a physical barrier. This barrier restricts the free movement of the electron, essentially trapping it within the particle's boundaries*

Quantum Confinement Effect

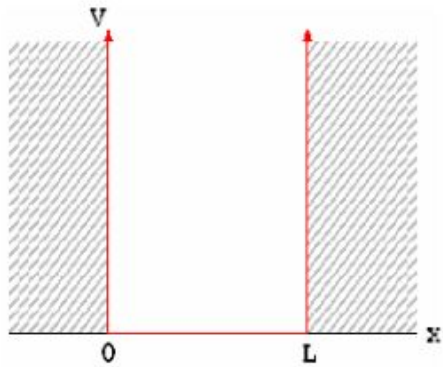
- **Restriction of Movement:** When the size of a particle becomes comparable to or smaller than the electron's wavelength, the particle acts as a physical barrier. This barrier restricts the free movement of the electron, essentially trapping it within the particle's boundaries.
- **Quantum Wells and Potential Wells:** This phenomenon creates a "potential well" for the electron, confining its motion to a limited space. If the particle is **confined in one dimension**, it is called a **quantum well**.

Why does the particle act like a barrier?

- A wave cannot "fit" into a space smaller than its wavelength without distortion.
- So the particle boundaries **restrict the wavefunction**, forcing it to satisfy boundary conditions (e.g., wavefunction = 0 at the walls).
- This means only certain standing wave patterns are allowed → **quantization of energy levels** (like particle in a box).
- To the electron, the finite-size particle looks like a **potential well (barrier)** because it cannot propagate beyond the particle freely

So, when a particle size $\leq$ electron wavelength, the electron is no longer free — it is confined and quantized. This is the principle behind quantum dots, nanocrystals, and semiconductor nanostructures.

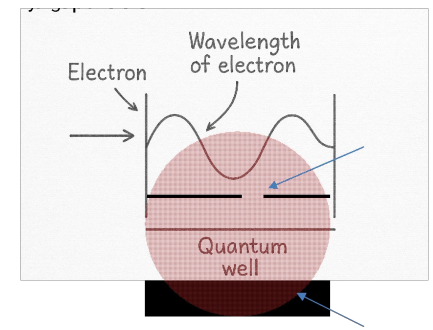
Quantum confinement in Particle in a 1D Box



Consider a particle of mass m confined to a 1D box of length L , with infinitely high walls at $x = 0$ and $x = L$.

- Inside the box: $V(x) = 0$
- Outside the box: $V(x) = \infty$

The particle cannot exist outside the box.



So the **time-independent Schrödinger equation** inside the box is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\psi(x) \quad \text{with boundary conditions}$$

$$\psi(0) = 0 \text{ and } \psi(L) = 0$$

The general solution is:

• Apply $\psi(0) = 0 \Rightarrow B = 0$

• Apply , where $n = 1, 2, 3, \dots$

So the normalized wavefunction is $\psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}, n = 1, 2, 3, \dots$

The energy is related to k by:

$$E = \frac{\hbar^2 k^2}{2m}$$

Substitute $k = \frac{n\pi}{L}$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, n = 1, 2, 3, \dots$$

Substitute $\hbar = \frac{h}{2\pi}$

$$E_n = \frac{n^2 h^2}{8mL^2}, n = 1, 2, 3, \dots$$

Discrete energy levels

Optical Properties of semiconductor nanomaterials

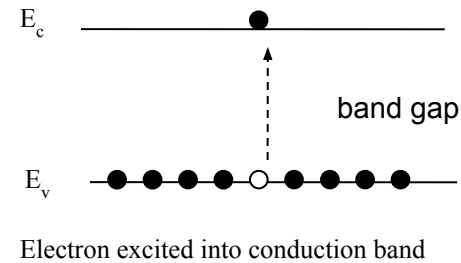
$$E_{g(qd)} = E_{\text{bulk}} + \frac{h^2}{8R^2} \left(\frac{1}{m_e^*} + \frac{1}{m_h^*} \right) - \frac{1.786e^2}{4\pi\epsilon_o\epsilon_r R^2}$$

Band gap of quantum dots (semiconductor particle that has size comparable to the wavelength of the electron)

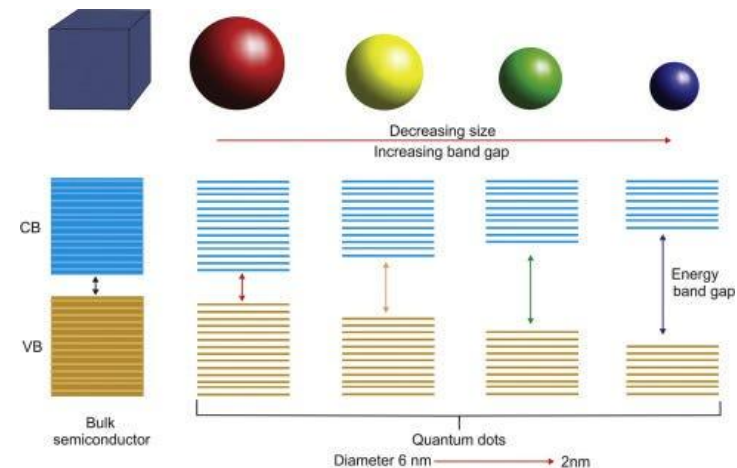
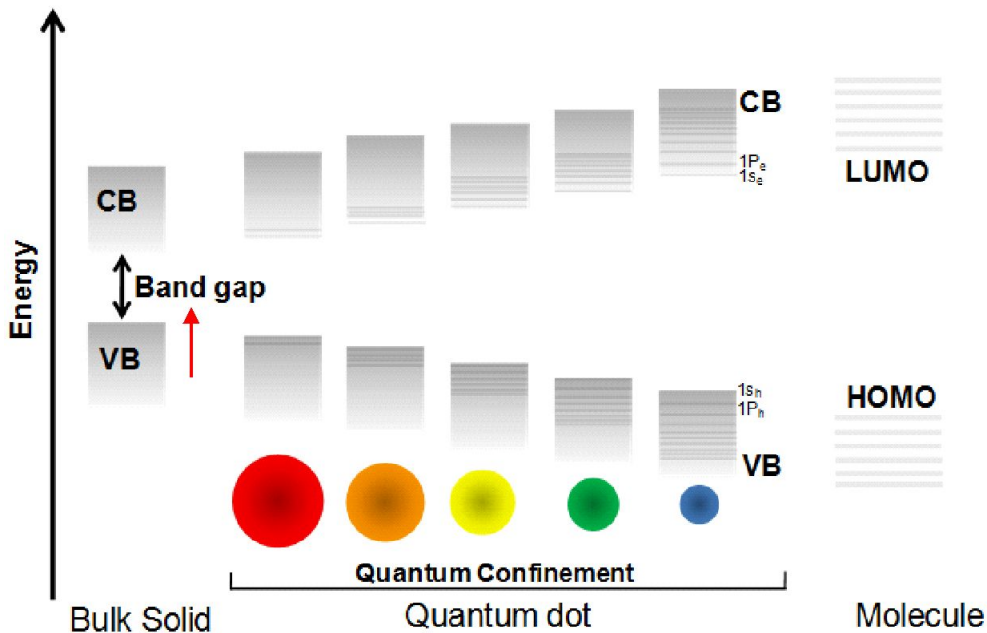
Confinement energy term

Coulomb energy term

(Brus equation)



In semiconductors, the **valence band** corresponds to the ground states of the valence electrons. In semiconductors, the **conduction band** corresponds to excited states where electrons are free to move about in the material and participate in conduction



• **Discrete Energy Levels:** In bulk materials, electrons occupy a continuous band of energy levels. However, in confined nanomaterials, the electrons can only exist in discrete energy levels, similar to [energy levels in an atom](#).

Consequences of Confinement

• **Increased Band Gap:** The restriction of electron motion leads to a larger energy gap (band gap) between the valence and conduction bands.

• **Size-Dependent Optical Properties:** A larger band gap means that the material will absorb and emit light at shorter wavelengths, a phenomenon known as the [blue shift](#). This size-dependent optical property is a hallmark of the quantum confinement effect.

What are Quantum Dots??

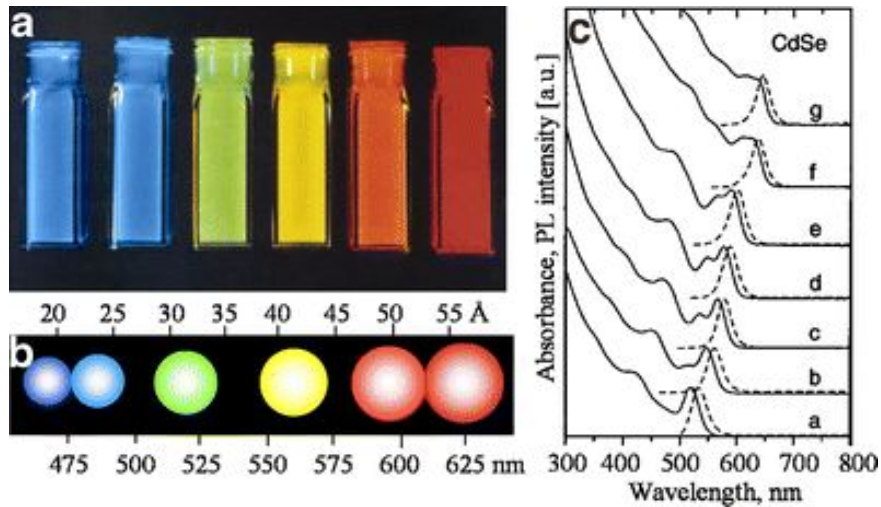
- A **quantum dot** is a semiconductor whose **excitons (bound-electron hole pair)** are **confined** in all three spatial dimensions. As a result, they have properties that are between those of bulk semiconductors and those of discrete molecules.
- A crystal of semiconductor compound (eg. CdSe, PbS) with a diameter on the order of the compound's Exciton Bohr Radius. Quantum dots are between 2 and 10 nanometers size.

What is excitonic Bohr radius?

*The **exciton Bohr radius (a_0)** is the characteristic, most probable separation distance between an [electron](#) and a [hole](#) that are bound together in an exciton, analogous to the Bohr radius of a hydrogen atom. In a [semiconductor](#), when the size of the quantum dot (QD) becomes comparable to or smaller than this exciton Bohr radius, the electrons and holes become spatially confined, leading to [quantum confinement effects](#) and altered optical and electronic properties, such as a [larger band gap](#).*

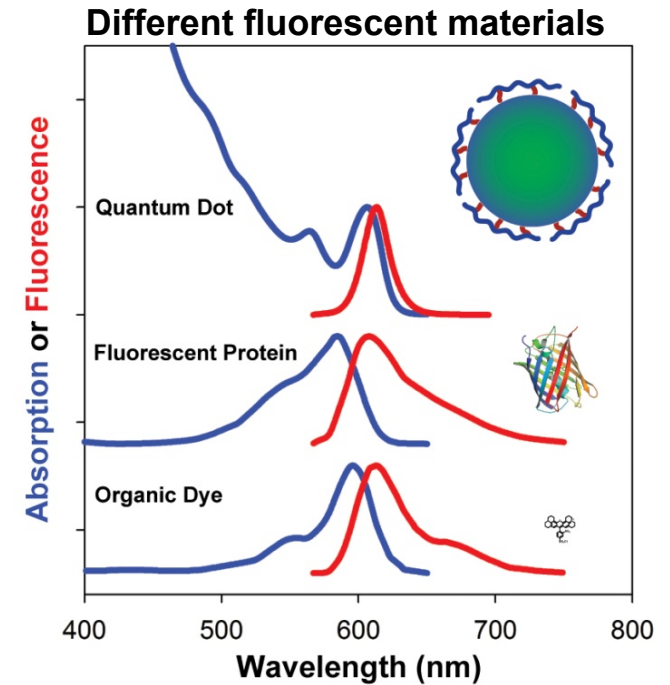
Quantum Confinement effect

Band gap tuning by tuning the size of quantum dots



QDs Compared with fluorescent dyes/proteins

- Brightness: 10-100 times higher
 - High molar absorption coefficient
 - Size-tunable: color
 - Multiplexing capabilities
 - narrow emission band, and wide excitation wavelength
 - Photostability: 100-1000 time higher
- Similar emission wavelength but different efficiency



Particle in a “box” : Quantum Dots

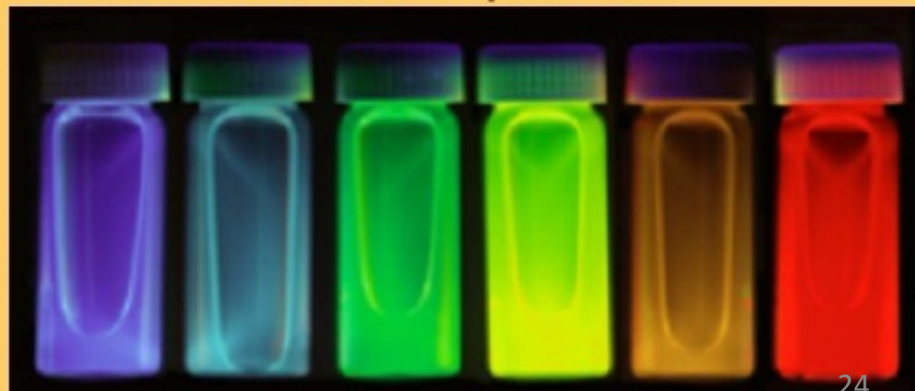
The color of light emitted by a semiconductor material is determined by the width of the energy gap separating the conduction and valence energy bands.

In bulk semiconductors, this gap width is fixed by the identity (*i.e.* composition) of the material. For example, the band gap energy of bulk CdSe is 1.77 eV at 300 K (infrared).

If you want a different wavelength of light to be emitted, you need to find a different material.

However, for nanoparticles the size of the particle imposes geometrical constraints on the electrons, which respond by adjusting their energy. This phenomenon is called the quantum size effect.

CdSe quantum dots emit in the visible region under UV illumination



Quantum Confinement

- Quantum confinement is observed when the size of the particle is too small to be comparable to the wavelength of electron.
- Confinement: to confine the motion of randomly moving electron, to restrict its motion in specific energy levels.
- Size of particle decreases till we reach a nano scale.
- The decrease in confining dimensions makes the energy level discrete and this increases the band gap and band gap energy also increases.
- Changes in atomic structure – due to the influence of small length scale in energy band structure
- Quantum confinement range for a semiconductor (1-25nm)
- Changes in particle size by adjusting the energy

Important Notes!

- Excitons have an average physical separation between the electron and hole, referred to as the Exciton Bohr Radius this physical distance is different for each material.
- In bulk, the dimensions of the semiconductor crystal are much larger than the Exciton Bohr Radius, allowing the exciton to extend to its natural limit.
- However, if the size of a semiconductor crystal becomes small enough that it approaches the size of the material's Exciton Bohr Radius, then the electron energy levels can no longer be treated as continuous - they must be treated as discrete, meaning that there is a small and finite separation between energy levels. This situation of discrete energy levels is called quantum confinement .

Why do we consider the hole mass in the reduced mass in the expression of band gap of quantum dots.

When we model the confinement of electrons and holes in a spherical quantum dot (QD), the lowest-order expression for the energy shift (**Brus equation**) is

$$E_{QD} = E_g + \frac{\hbar^2 \pi^2}{2R^2} \left(\frac{1}{m_e^*} + \frac{1}{m_h^*} \right) - \frac{1.8e^2}{4\pi\epsilon R}$$

Why the hole mass enters?

- In semiconductors, an **excited electron in the conduction band** always comes with a **hole** left behind in the valence band.
- They form an **electron–hole pair (exciton)**, bound by Coulomb attraction.
- **Quantum confinement doesn't affect the electron alone — it confines the center-of-mass motion of the exciton** inside the quantum dot

- E_g : bulk band gap of the semiconductor
- R : quantum dot radius
- m_e^* : effective electron mass
- m_h^* : effective hole mass
- ϵ : dielectric constant

The relative motion of the electron and hole is treated like a **two-body problem** in quantum mechanics.

In quantum mechanics

$$\mu = \frac{m_e^* m_h^*}{m_e^* + m_h^*}$$

This **reduced mass** μ describes the relative motion of two particles under confinement.

The reduced mass, which is a combination of the electron's and the hole's effective masses, simplifies the two-body problem of the electron and hole into a one-body problem, making it easier to model the quantized energy levels and the resulting size-dependent band gap in the quantum dot.

Can gold behave as semiconductor?

The electrical properties of nanomaterials mainly concern the mobility of the charge carriers. When the dimensions of a material are reduced to the nanometer range, the quantum-size effect and quantum confinement effect are bound to occur. Because of the quantization of electron energy, some conductive metal nanoparticles can be turned into nonconductive under certain voltage. For example, metals used as conductors such as copper can lose their conductivity if their size is reduced to a few nanometers.

Increased interfacial atoms or ions, sinking a large amount of defects near the grain boundaries, grain size, and doping impurities largely controls the dielectric properties, resistivity and electrical transports of nanomaterials. The electrical conductivity of nanomaterials is generally lower than the bulk materials due to the increase of the band gap energy with a decrease in particle size of the nanomaterials, especially for semiconductor nanomaterials. *For example, as the energy bands cease to overlap at 2–3 nm range, in spite of being metallic, gold nanoparticles cease to be conductive and turn into insulators.*

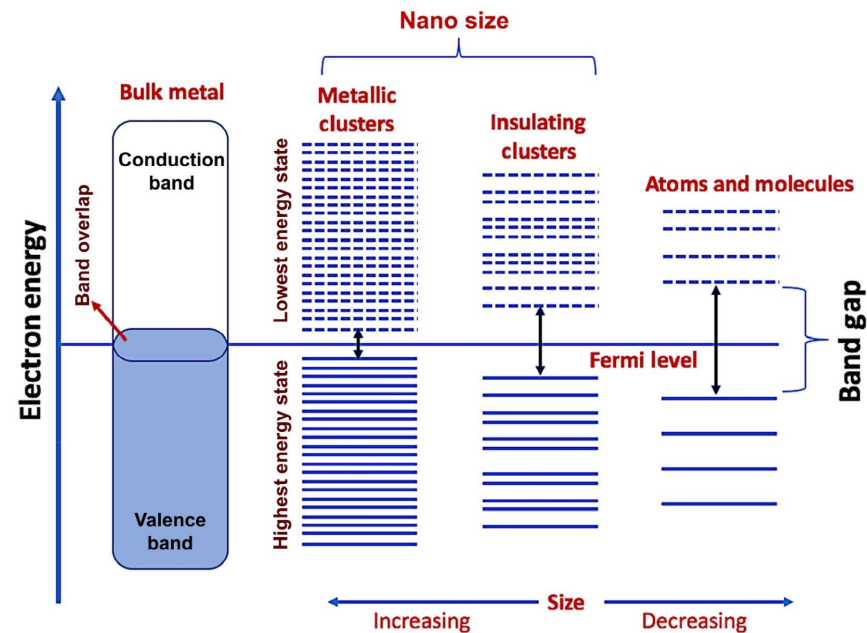
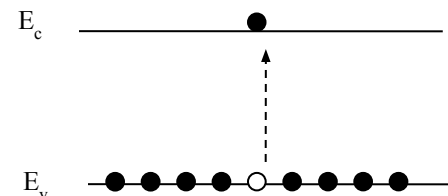
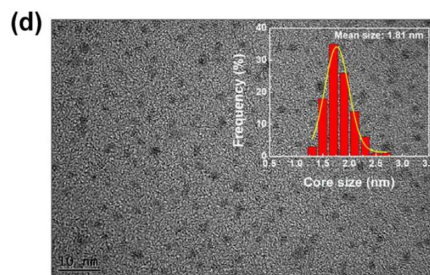
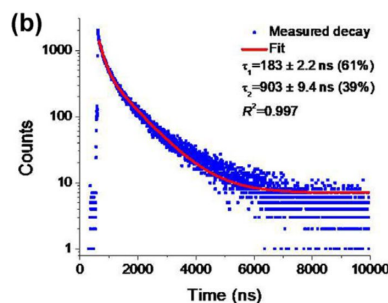
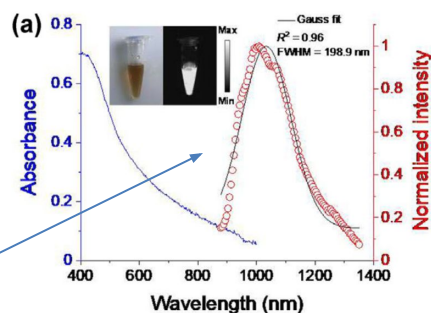


FIG. 15.10 Size induced metal-insulator transition. Reproduced from reference [48].

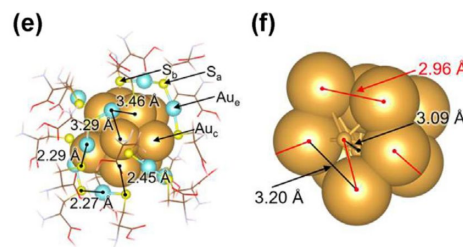
Gold Nanocluster behave as semiconductor

Nanoclusters are atomically precise, crystalline materials most often existing on the 0-2 nanometer scale. They are often considered kinetically stable intermediates that form during the synthesis of comparatively larger materials such as semiconductor and metallic nanocrystals.

Emission from
gold nanocluster



Electron excited into conduction band



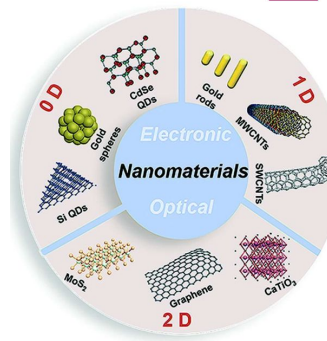
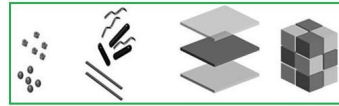
(a) Absorption (blue line) and normalized fluorescence (red line) spectra of Min-23@AuNCs ($\lambda_{\text{exc}} = 808 \text{ nm}$).

(b) Fluorescence decay curve of Min 23@AuNCs dispersed in water

(d) TEM image of AuNCs. Inset shows histogram of particle size distribution (core size, $d = 1.81 \pm 0.26 \text{ nm}$; polydispersity index, $\text{PDI} = 0.02$). Optimized structure of (e) the $\text{Au}_{25}(\text{SCH}_2\text{CHNH}_2\text{COOH})_{18}$ cluster and (f) icosahedral Au_{13} core.

Classification of nanomaterials based on Dimensionality and quantum confinement

- zero-dimensional (0D),
- one-dimensional (1D),
- two-dimensional (2D)
- **three-dimensional (3D) nanomaterials:**



(i) Zero-dimensional nanomaterials: Here, all dimensions (x, y, z) are at nanoscale, i.e., no dimensions are greater than 100 nm. It includes nanospheres and nanoclusters.

(ii) One-dimensional nanomaterials: Here, two dimensions (x, y) are at nanoscale and the other is outside the nanoscale. This leads to needle shaped nanomaterials. It includes nanofibres, nanotubes, nanorods, and nanowires.

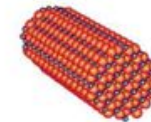
(iii) Two-dimensional nanomaterials: Here, one dimension (x) is at nanoscale and the other two are outside the nanoscale. The 2D nanomaterials exhibit plate-like shapes. It includes nanofilms, nanolayers and nanocoatings with nanometre thickness.

(iv) Three-dimensional nanomaterials: These are the nanomaterials that are not confined to the nanoscale in any dimension. These materials have three arbitrary dimensions above 100 nm. The bulk (3D) nanomaterials are composed of a multiple arrangement of nanosize crystals in different orientations. It includes dispersions of nanoparticles, bundles of nanowires and nanotubes as well as multi-nanolayers (polycrystals) in which the 0D, 1D and 2D structural elements are in close contact with each other and form interfaces.

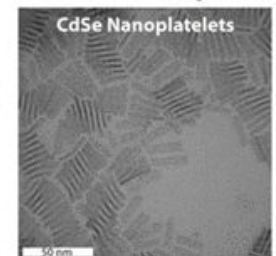
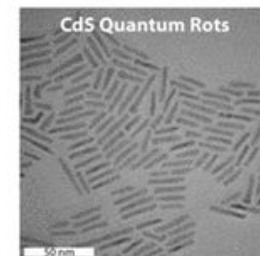
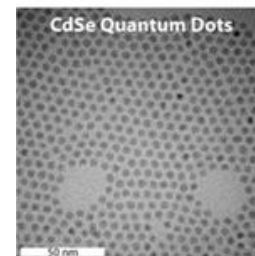
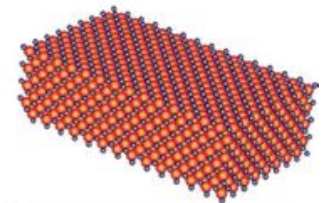
3D Confinement



2D Confinement



1D Confinement



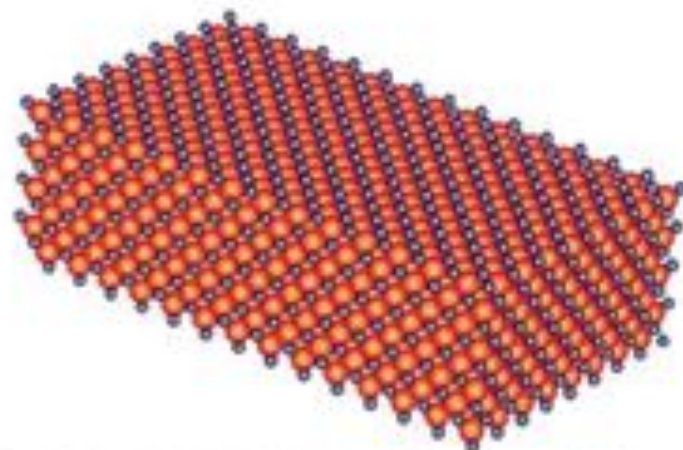
3D Confinement



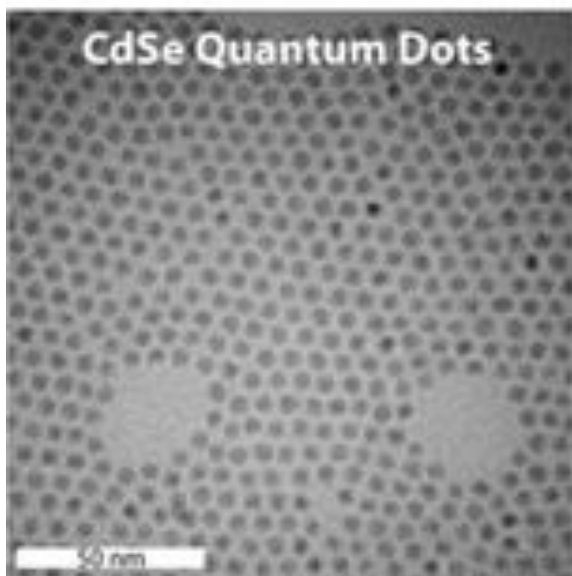
2D Confinement



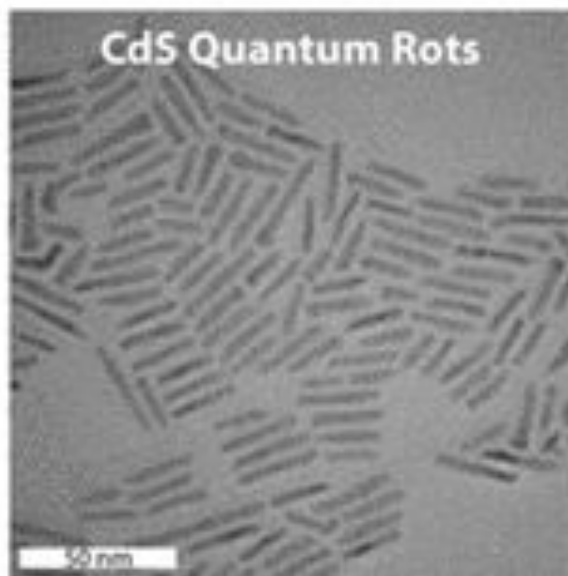
1D Confinement



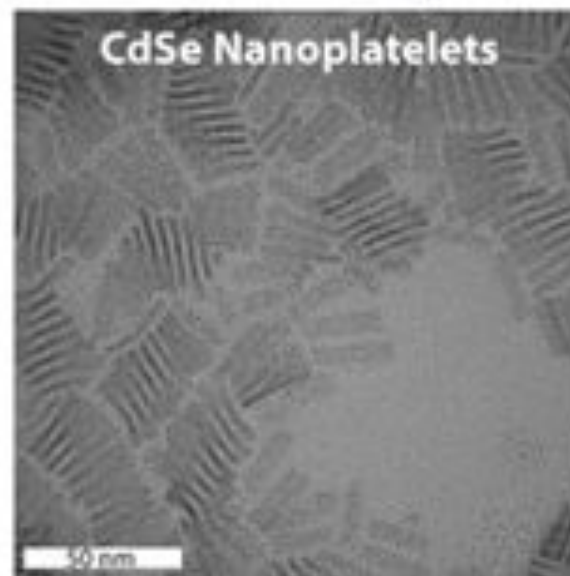
CdSe Quantum Dots



CdS Quantum Rods



CdSe Nanoplatelets



Quantum Confinement

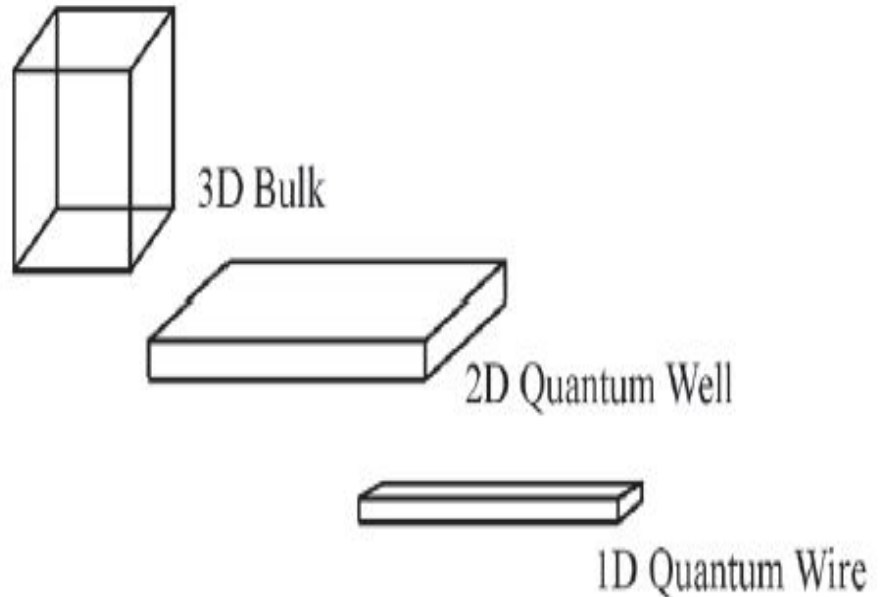
3-D: All carriers act as free carriers in all three directions

2-D or Quantum Wells: The carriers act as free carriers in a plane

First observed in semiconductor systems

1-D or Quantum Wires: The carriers are free to move down the direction of the wire

0-D or Quantum Dots: Systems in which carriers are confined in all directions (no free carriers)



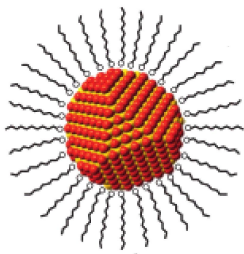
3D Confinement



Discussions on 0D nanomaterials

•Dimensionality Classification:

- Nanomaterials are classified based on the number of dimensions that are in the nanoscale range (typically 1-100 nm). A 0D nanomaterial has all three dimensions in the nanoscale.



Zero-dimensional (0D) nanoparticles, also known as [quantum dots](#), are nanomaterials where all three spatial dimensions are within the nanoscale, typically less than 100 nm. They exhibit unique properties due to quantum mechanical effects and high [surface-to-volume ratio](#).

•Examples of 0D materials:

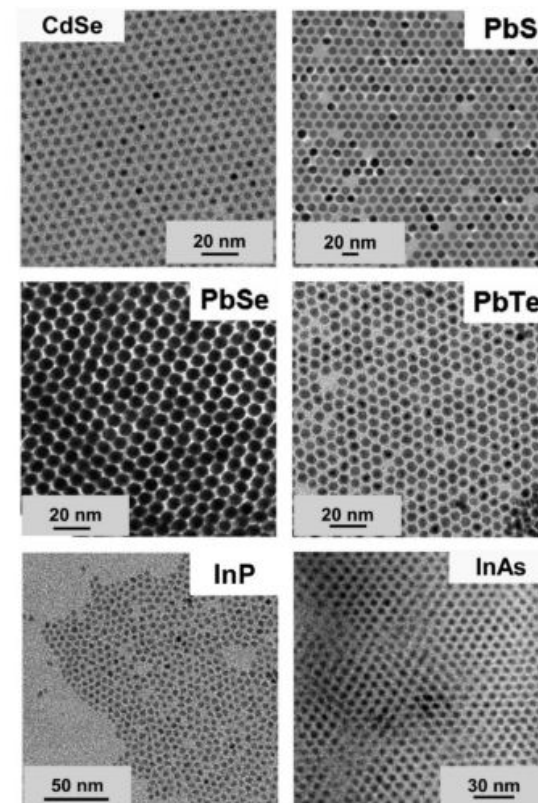
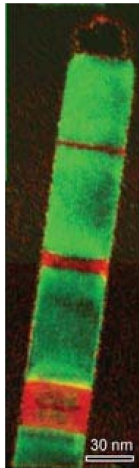


Figure 2. Examples of colloidal semiconductor nanocrystals of different materials.

A nanowire is a nanostructure with a diameter typically between 1 and 100 nanometers and a length that can be much longer, often up to tens of micrometers. These one-dimensional structures exhibit unique mechanical, electrical, thermal, and optical properties, making them promising materials for nanoscale electronic and optoelectronic devices.

Q: What is a nanowire?

- A: Any solid material in the form of wire with diameter smaller than about 100 nm



Transmission electron micrograph of an InP/InAs nanowire
(M.T. Bjork et. al., Nanoletters, 2:2 2002)

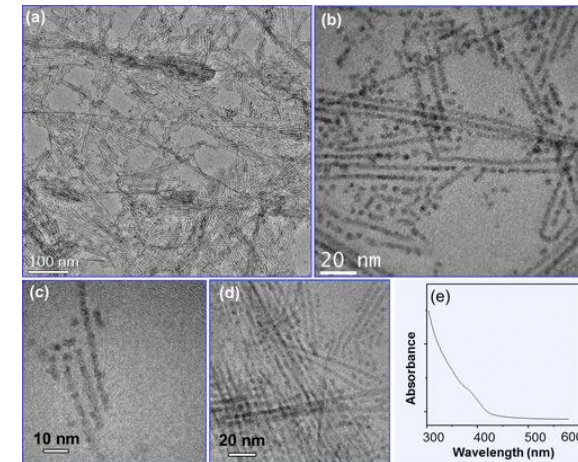
- **Dimensions:**

Nanowires are defined by their high aspect ratio, meaning their length is significantly greater than their diameter (typically a ratio of 1000:1 or more).

- **Material Variety:**

They can be made from various materials, including metals (e.g., Ni, Au, Pt), semiconductors (e.g., Si, GaN, InP), and insulators (e.g., SiO₂, TiO₂).

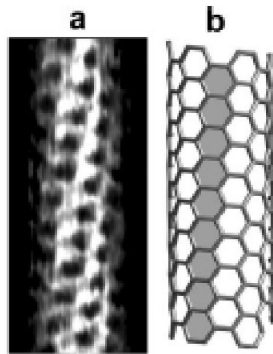
Example of ZnSe nanowires



The Journal of Physical Chemistry Letters 4, 3292-3297

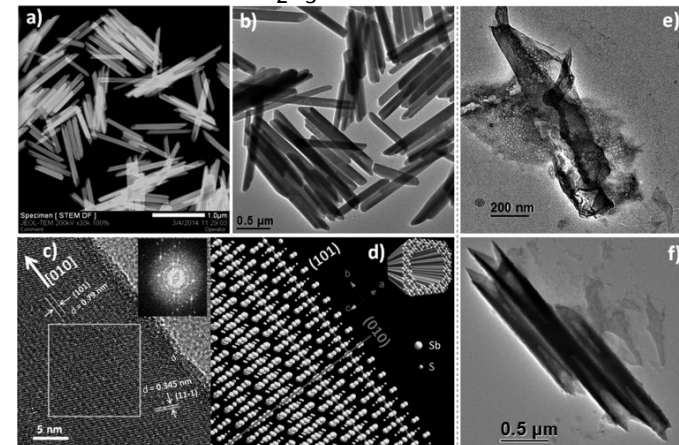
Q: What is a nanotube?

- A: A hollow nanowire, typically with a wall thickness on the order of molecular dimensions
- The smallest (and most interesting) nanotube is the single-walled carbon nanotube (SWCNT) consisting of a single graphene sheet rolled up into a tube



Scanning
Tunneling
Micrograph of a
single-walled
carbon **nanotube**
and corresponding
model (Dekker)

Sb_2S_3 nanotube



<https://doi.org/10.1002/anie.201405148>

A nanotube is a tube-like structure with a diameter on the nanometer scale (10^{-9} meters). Carbon nanotubes (CNTs) are a prominent type, composed of rolled-up [graphene](#) sheets, known for their exceptional strength, electrical conductivity, and thermal properties. They can be single-walled (SWCNT) or multi-walled (MWCNT).

Other Nanotubes:

Besides carbon, nanotubes can be made from materials like boron nitride, gallium nitride, antimony sulphide, silicon.

Q: What makes nanowires and nanotubes (scientifically) interesting?

- **Electronic & optical properties**
 - Nanowires and nanotubes are the most confining electrical conductors - puts the squeeze on electrons
 - Can be defect free - electrons move “ballistically”
 - Quantum confinement - tunable optical properties
- **Mechanical properties**
 - Small enough to be defect-free, thus exhibiting ideal strength
- **Thermal properties**
 - Can be designed to conduct heat substantially better (or much worse) than nearly every bulk material
- **Chemical properties**
 - Dominated by large surface-to-volume ratio

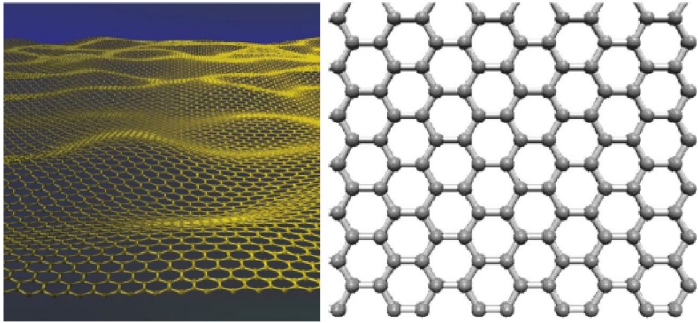
• **Applications:**

Nanowires are being explored for a wide range of applications, including:

- **Electronics:** Field-effect transistors (FETs), diodes, and logic gates.
- **Optoelectronics:** Sensors, solar cells, and light-emitting devices.
- **Energy Storage:** Nanowire-based batteries.
- **Sensors:** Highly sensitive and real-time sensors for biological and chemical species.
- **Healthcare:** Targeted drug delivery, nerve stimulation, and toxin detection

Colloidal quantum wells (CQWs), or nanoplatelets, nanosheet

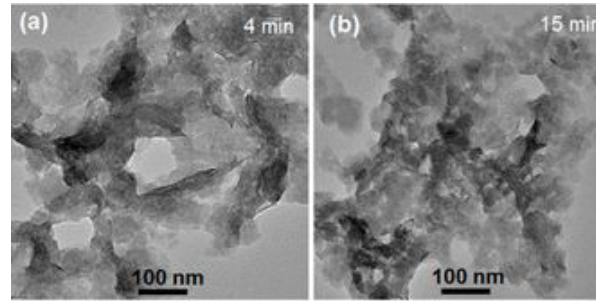
A nanosheet is a two-dimensional nanostructure with thickness in a scale ranging from 1 to 100 nm. A typical example of a nanosheet is graphene, the thinnest two-dimensional material (0.34 nm) in the world. It consists of a single layer of carbon atoms with hexagonal lattices



Graphene

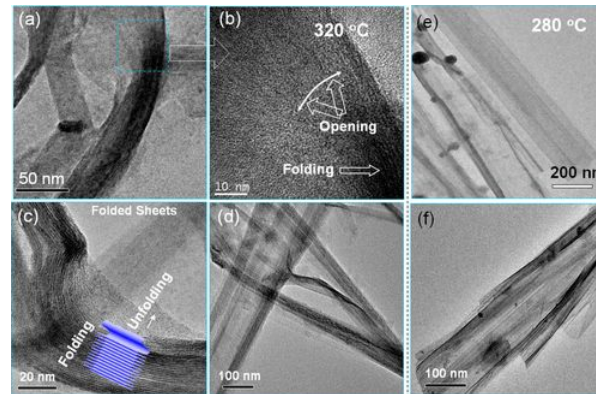
The Nobel prize in Physics 2010

Andre Geim, Konstantin Novoselov



In_2S_3 nanosheet

Chem.Mater.2016, 28,
4324–4330

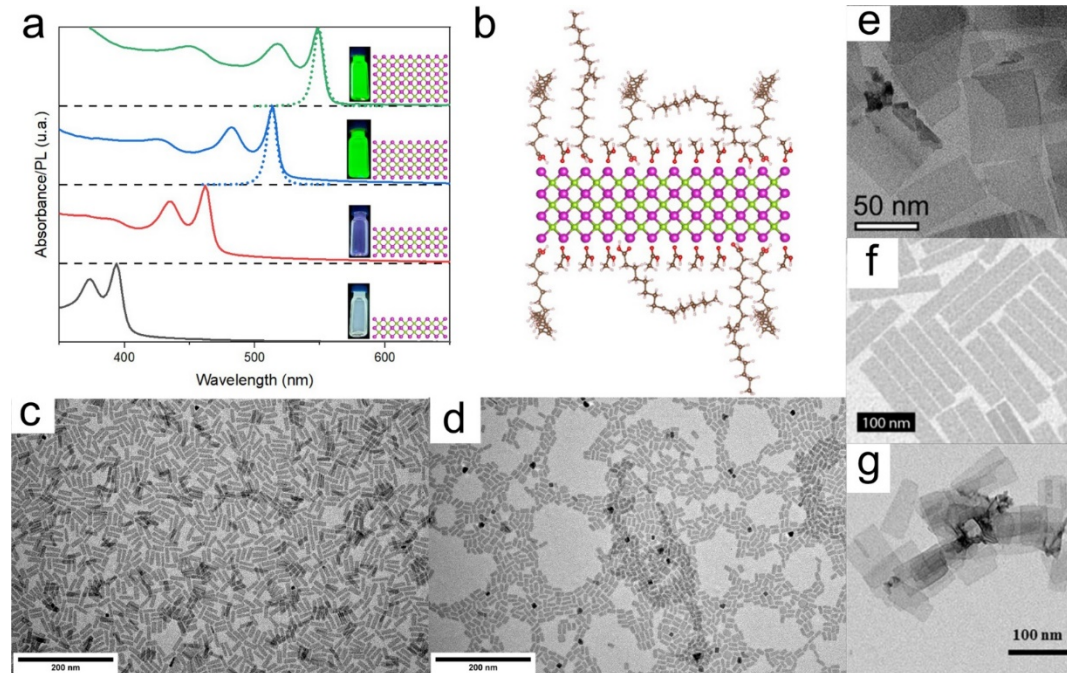


J. Phys. Chem. Lett. 2012, 3, 24, 3812–3817

In_2S_3 nanosheet

Optical Properties, surface ligands, and electron microscopic images of flat semiconducting 2 dimensional CdSe nanoplatelet

Thickness dependent emission and absorption

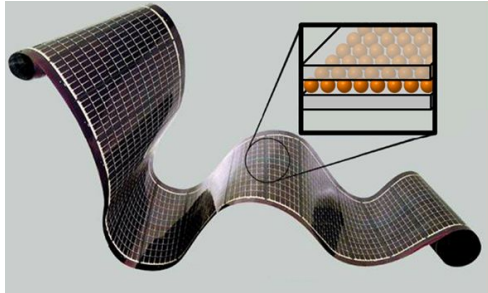


CdSe nanoplate

a UV-vis absorbance (full line) and fluorescence (dotted line) spectra of 2 to 5 ML CdSe NPLs (from bottom to top). **b** Schematic representation of the cross-section of a 3 ML CdSe NPL with oleic acid and acetate ligands on its surface. Magenta corresponds to Cd, green to Se, red to O, brown to C, and white to H. **c–g** TEM images of **c** 4 ML CdSe NPLs, **d** 5 ML CdSe NPLs, and **e** 4 ML CdSe NPLs with a monolayer of CdS. **f** 1.2 nm-thick PbS nanosheets. **g** 2.2 nm-thick CdTe NPLs

QDs in various applications

Solar cell



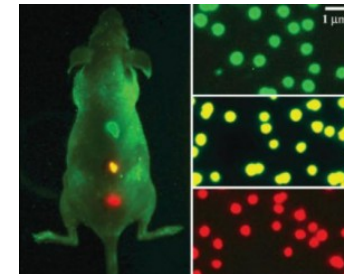
Energy harvesting

Television



Display device

Bio-imaging



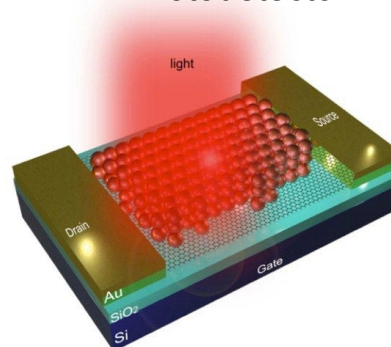
Contrasting agent

LEDs



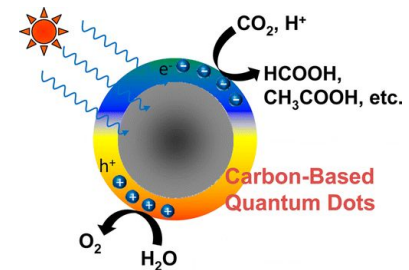
Lighting

Photodetector



Photon sensing

Photocatalyst

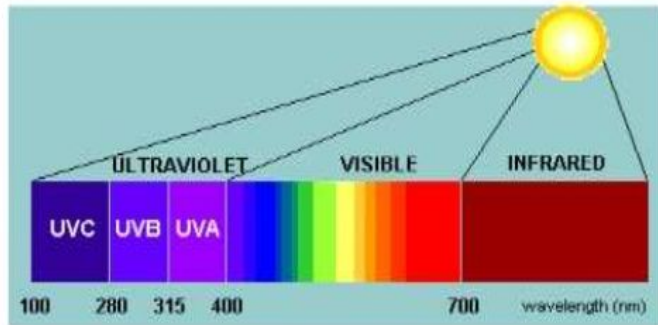


catalyzing chemical reaction

Applications

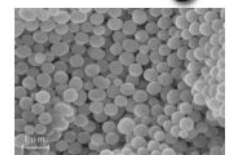
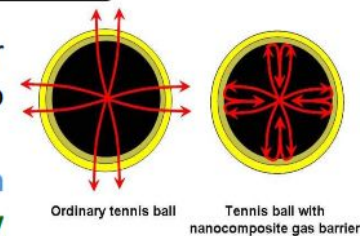
Cosmetics

- Zinc oxide and titanium oxide are both employed as opaque sunblocks.



Sports materials

- **Polymer nanocomposites** barrier film prevents air loss from **Tennis balls (InMat)**, so that tennis ball do not lose its bounce.
- **Tennis racket (Wilson)** frames containing **silicon dioxide nanoparticles** - Increases **strength, stability and power**.
- **Resin containing buckyballs** (fullerenes) used to make **badminton raquets (Yonex)**.
- **Bicycle (Eston Cycling)** parts made with **carbon nanotubes**- Increases stiffness without weight increase.
- **Fishing rods (St Croix Rods)** made with a **epoxy resin in silica nanoparticles**- Increased strength without weight increase.
- Jeremy Wariner used a new **shoe** designed using nanotechnology by **Adidas** (in the event men's 400m race) **in the 2008 Olympic Games**.





Next-generation computer chips

Ultra-high purity materials, enhanced thermal conductivity and longer lasting nanocrystalline materials

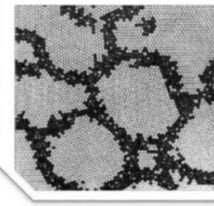


Improved HDTV and LCD monitors

Nanocrystalline selenide, zinc sulfide, cadmium sulfide, and lead telluride to replace current phosphors
Cheaper and more durable.

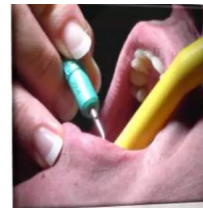
Elimination of Pollutants

- Engineered to be chemically reactive to carbon monoxide and nitrous oxide
- More efficient pollution controls and cleanup



TOUGHER AND HARDER CUTTING TOOLS

- Tungsten carbide, tantalum carbide, and titanium carbide
- Much more wear-resistant and corrosion-resistant than conventional materials
- Reduces time needed to manufacture parts, cheaper manufacturing



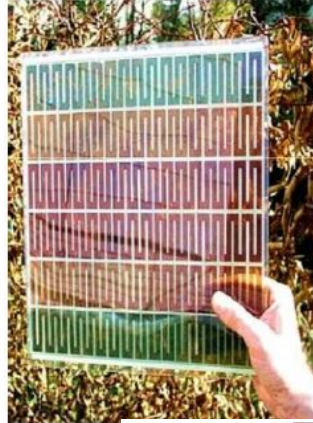
LONG LASTING MEDICAL IMPLANTS

- Higher quality medical implants
 - Current micro-scale implants aren't porous enough for tissue to penetrate and adapt to
 - Nano-scale materials not only enhance durability and strength of implants but also allow tissue cells to adapt more readily



Energy sector

- Fuel cell catalysis; hydrogen production photo-catalysis; solar cells; lithium ion battery etc.
- Newer solar panels now incorporate “nanocrystalline silicon”. **This increases conversion efficiency.**
- Other Examples: Titanium Dioxide Nanoparticles in Dye Solar Cells; Fullerene Derivates as Electron Acceptors in Polymer Solar Cells; Thin-Layer Solar Cells etc.



Nanomaterials for Aviation Industry

• Properties of Materials required for Airframe Structure:

Light Weight; High Strength; High Toughness; Corrosion Resistance; Easy Reparability & Reusability; Less Maintenance & Durability .

Nanomaterials can fulfill these requirements.

- Some **CNT based composites** which can be used for airframe structure are - CNT/Epoxy, CNT/Polyimide: **wide range of Young's Modulus, High Specific Strength and Thermal Performance.**

- **Nanoclays** reinforced Polymer Composites: **Barrier Properties, Thermal and Flame Retardant**

- **Metal Nps** incorporated Composites: **extraordinary electrostatic discharge and electromagnetic interference (EMI) shielding properties**

- **SiC Nanoparticles** in alumina Yittria stabilized nanozirconia: **crack healing, resulting in improved high-temperature, strength and creep resistance.**



Nanotechnology in Electronics

Advantages of Using Nanotechnology in Electronics:

- **Reducing the size of transistors** used in integrated circuits & increasing the density of memory chips
- **Decreasing the weight and thickness of the screens**
- **Improving display screens** on electronics devices.
- **Reducing power consumption.**

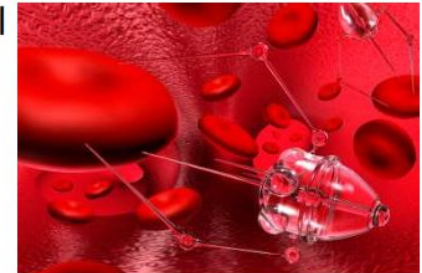
Displays

Carbon nanotubes on a glass or plastic sheet allow manufacturers to make clear conductive panels for displays that are extremely thin.



Nano-medicine

- There are many possible medical uses of nanotechnology.
- **Nano-coatings** on hip and joint replacements to prevent rejection.
- **Nano-capsules of drugs** that will target cancer cells only i.e. nanotechnology may also be useful for developing ways to eradicate cancer cells without harming healthy, neighbouring cells.
- Detection and diagnosis, biomarking and imaging, MRI contrast agents, drug delivery, dental ceramics etc.
- **Plasters and bandages** can contain nanocrystals of **silver**, because it is toxic to bacteria.



Food and Agriculture

- **Nano-Pesticides, Nanosensors** for soil quality and for plant health monitoring
- **Food packaging** : Silicate nanoparticles in plastic packaging can provide a barrier to gases (oxygen) or moisture
- **UV-protection** : Zinc Oxide nanoparticles can block UV rays and provide anti-bacterial protection when added to plastic packaging
- **Protection from microorganism**: Nano-Ag (Silver nano refrigerator) etc.



Nano-cellulose food packaging

Environment

- **Pollution control and monitoring** :
 - ✓ Ceria (CeO_2) nanocrystals doped with Au- nanoparticles can catalyze the oxidation of CO to CO_2 at a much lower temperature- this can be used in automobiles to reduced air pollution.
 - ✓ TiO_2 nanoparticle-based photocatalytic degradation of air pollutants in self-cleaning systems
 - ✓ Nanocatalysts for more efficient, and better-controlled catalytic converters
 - ✓ Nanosensors for detection of toxic materials and leaks
 - Gas separation nanodevices.

Water treatment and remediation:

- **Nanosensors** for the **detection of contaminants** and pathogens;

Nanoporous zeolites, nanoporous polymers, and attapulgite clays for water purification;

Magnetic nanoparticles for water treatment and remediation;

TiO₂ nanoparticles for the **catalytic degradation** of water pollutants.

Nanomembranes for **water purification**, desalination, and detoxification.

Nanotechnology spans other Areas

- **Industrial:**
 - Scratch-Resistant and Dirt-Resistant **Paints**; optical engineering, Nanoparticle Fillers for **Tyres** (enhance tyre performance and durability), catalysis, fabrication of **cutting tools and hard tip** etc.
 - Samsung says that "Samsung WM1245A **Washing Machine** releases over 400 billion silver ions which penetrate deeply into fabrics of any kind and create a coat of sterilizing protection for a maximum of 99.99% disinfection and an added antibacterial effect of up to 30 days after washing".
 - **Defence and security:** Highly sensitive sensors, smart materials etc.
 - And many more.....



Synthesis Techniques of quantum dots

